NEET UG PHYSICS PREPARATION

Gravitation

Comprehensive Theory, Derivations, Worked Examples, and NEET UG Practice Questions

1. Learning Objectives & Historical Overview

Learning Objectives:

- Understand the historical transition from the geocentric model to the heliocentric model of planetary motion.
- State and apply Kepler's three laws of planetary motion and relate the law of areas to angular momentum conservation.
- Formulate Newton's Universal Law of Gravitation in scalar and vector forms and apply the principle of superposition.
- Understand the shell theorems and Cavendish's experimental determination of the universal gravitational constant G.
- Derive and calculate the variation of acceleration due to gravity g with altitude h and depth d.
- Calculate gravitational potential energy V and gravitational potential U for discrete and continuous mass distributions.
- Derive escape speed v_e and analyze the orbital velocity, time period, and mechanical energy of Earth satellites.

1.1 Historical Development of Gravitation

From early life, we observe that all material objects are attracted towards the Earth: objects thrown up fall down, climbing uphill is more tiring than walking downhill, and raindrops fall towards the ground. Historically, the Italian physicist **Galileo Galilei (1564–1642)** recognized that all bodies, irrespective of their masses, are accelerated towards the Earth with a constant acceleration. Through experiments with bodies rolling down inclined planes, Galileo arrived at a value of the acceleration due to gravity very close to the modern accurate value.

Alongside terrestrial observations, the motion of stars and planets against the fixed background of stars fascinated early astronomers:

- **Geocentric Model (Ptolemy, ~2000 years ago):** Proposed that all celestial objects—stars, the Sun, and the planets—revolved around a stationary Earth in circular motions (with planets moving in circles whose centers moved in larger circles).
- **Heliocentric Concept in India (Aryabhatta, 5th Century A.D.):** Mentioned an elegant heliocentric model in his treatise where the Sun was the center around which planets revolved.
- **Copernican Model (Nicolaus Copernicus, 1473–1543):** A thousand years after Aryabhatta, Copernicus proposed a definitive model in which planets moved in circles around a fixed central Sun. Galileo strongly supported this model despite facing state prosecution.
- **Observational Foundation (Tycho Brahe, 1546–1601 & Johannes Kepler, 1571–1640):** Danish nobleman Tycho Brahe spent his lifetime recording naked-eye planetary observations. His assistant Johannes Kepler analyzed this compiled data and extracted three fundamental laws of planetary motion, which later enabled Isaac Newton to formulate the Universal Law of Gravitation.

2. Core Concepts & Detailed Explanations

2.1 Kepler's Laws of Planetary Motion

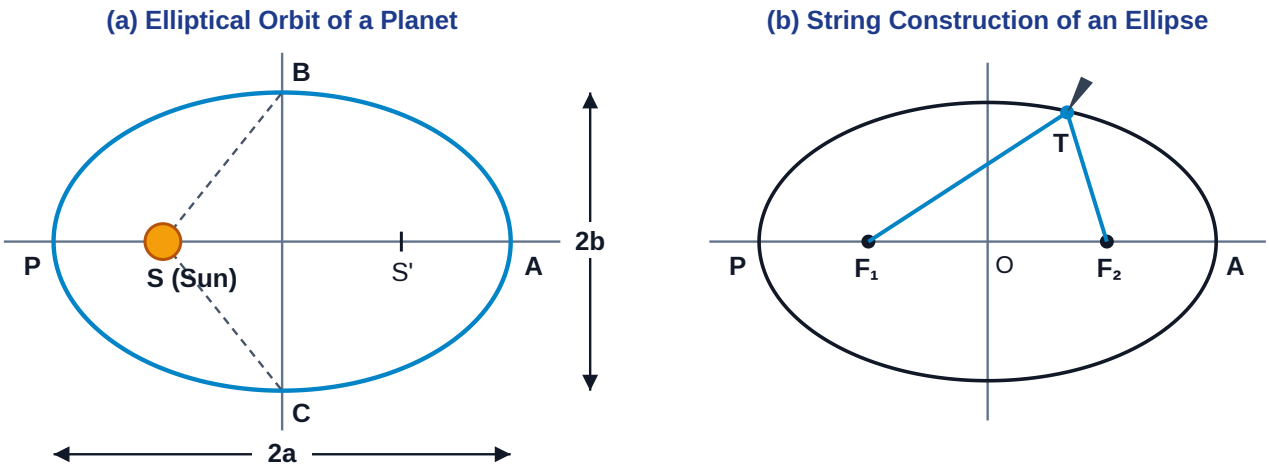
Johannes Kepler formulated three laws that govern the motion of planets around the Sun (and apply equally to satellites orbiting any central body):

1. Law of Orbits (First Law)

All planets move in elliptical orbits with the Sun situated at one of the foci of the ellipse. This was a major departure from the Copernican model, which permitted only circular orbits.

An ellipse is a closed curve that can be drawn by fixing the ends of a taut string at two points F_1 and F_2 (called the **foci**) and tracing a curve with a pencil keeping the string taut. For any point T on the ellipse, the sum of distances $TF_1 + TF_2$ is constant. The line passing through F_1 and F_2 intersects the ellipse at points P and A. The midpoint O of segment PA is the center of the ellipse, and the length $PO = AO = a$ is the **semi-major axis**. A circle is a special case of an ellipse in which the two foci merge into one at the center, and the semi-major axis becomes the radius of the circle.

- **Perihelion (P):** The point on the elliptical orbit closest to the Sun.
- **Aphelion (A):** The point on the elliptical orbit farthest from the Sun.



Tutorial Diagram 1: (a) Elliptical orbit of a planet around the Sun S with perihelion P, aphelion A, major axis 2a, and minor axis 2b. (b) Taut string construction showing constant sum of focal distances $TF_1 + TF_2$.

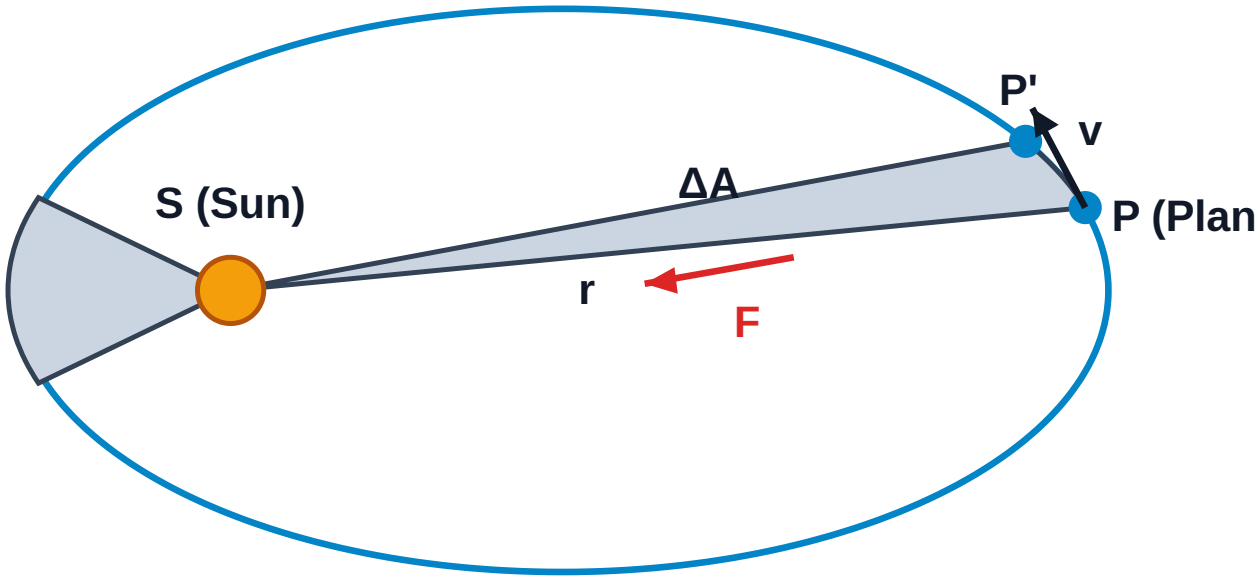
2. Law of Areas (Second Law)

The line that joins any planet to the Sun sweeps out equal areas in equal intervals of time. This law reflects the observation that planets move faster when they are nearer to the Sun (near perihelion) and slower when they are farther from the Sun (near aphelion).

Connection to Conservation of Angular Momentum: The law of areas is a direct consequence of conservation of angular momentum, which holds for any *central force* (a force directed along the vector joining the Sun and the planet). Let the Sun be at the origin and let the position vector, velocity, and linear momentum of a planet of mass m be $\mathbf{r}$, $\mathbf{v}$, and $\mathbf{p} = m\mathbf{v}$, respectively. The area $\Delta\mathbf{A}$ swept out by the radius vector in a small time interval Δt is:

$$\Delta\mathbf{A} = \frac{1}{2}(\mathbf{r} \times \mathbf{v}\Delta t)$$
$$\frac{\Delta\mathbf{A}}{\Delta t} = \frac{1}{2}\frac{\mathbf{r} \times \mathbf{p}}{m} = \frac{\mathbf{L}}{2m}$$

where $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is the angular momentum of the planet about the Sun. Because the gravitational force is a central force directed along $-\mathbf{r}$, the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{0}$, so $\mathbf{L}$ remains constant throughout the orbit. Consequently, the areal velocity $\frac{\Delta\mathbf{A}}{\Delta t} = \frac{\mathbf{L}}{2m}$ is constant.



Tutorial Diagram 2: Kepler's Second Law (Law of Areas). A planet P moves to P' in time interval Δt, sweeping out shaded area ΔA. Equal areas are swept in equal time intervals.

3. Law of Periods (Third Law)

The square of the time period of revolution (T) of a planet around the Sun is directly proportional to the cube of the semi-major axis (a) of its elliptical orbit:

$$T^2 \propto a^3 \implies \frac{T^2}{a^3} = \text{constant}$$

The following planetary measurement data confirm Kepler's Law of Periods across the eight planets of our Solar System:

Planet	Semi-major Axis a ($10^{10}\,\mathrm{m}$)	Time Period T (years, y)	Quotient $Q = T^2/a^3$ ($10^{-34}\,\mathrm{y^2\,m^{-3}}$)
Mercury	5.79	0.24	2.95
Venus	10.8	0.615	3.00
Earth	15.0	1.00	2.96
Mars	22.8	1.88	2.98
Jupiter	77.8	11.9	3.01
Saturn	143	29.5	2.98
Uranus	287	84.0	2.98
Neptune	450	165	2.99

2.2 Universal Law of Gravitation

Newton reasoned that the Moon revolving in a nearly circular orbit of radius $R_m \approx 3.84 \times 10^8\,\mathrm{m}$ with a time period $T \approx 27.3\,\mathrm{days}$ experiences a centripetal acceleration due to Earth's gravity given by:

$$a_m = \frac{V^2}{R_m} = \frac{4\pi^2 R_m}{T^2}$$

Comparing this acceleration with the acceleration due to gravity $g = 9.8\,\mathrm{m\,s^{-2}}$ at the Earth's surface (distance R_E from Earth's center), Newton showed that gravitational attraction decreases as the inverse square of distance from the center of the Earth:

$$\frac{g}{a_m} = \frac{R_m^2}{R_E^2} \simeq 3600$$

> **Newton's Universal Law of Gravitation:**
> Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

For two point masses m_1 and m_2 separated by a distance r, the magnitude of the gravitational force is:

$$|\mathbf{F}| = G\frac{m_1 m_2}{r^2}$$

Vector Form: Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be the position vectors of point masses m_1 and m_2, and let $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ be the displacement vector from m_1 to m_2 with unit vector $\hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|}$. Because the gravitational force is attractive, the force $\mathbf{F}$ on m_2 due to m_1 is directed along $-\hat{\mathbf{r}}$:

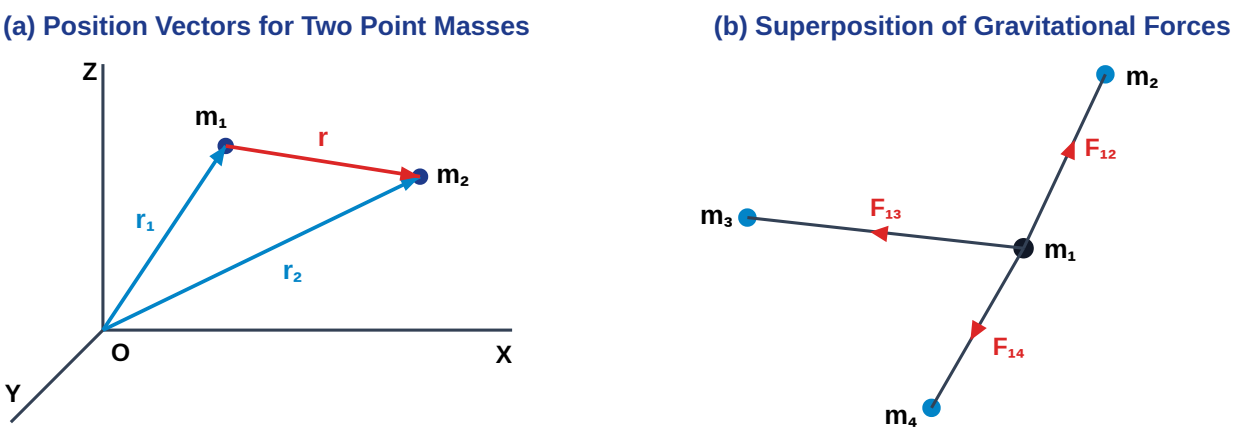
$$\mathbf{F} = G\frac{m_1 m_2}{r^2}(-\hat{\mathbf{r}}) = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}} = -G\frac{m_1 m_2}{|\mathbf{r}|^3}\mathbf{r}$$

By Newton's third law, the gravitational force $\mathbf{F}_{12}$ on body 1 due to body 2 and $\mathbf{F}_{21}$ on body 2 due to body 1 satisfy $\mathbf{F}_{12} = -\mathbf{F}_{21}$.

Principle of Superposition of Gravitational Forces

For a collection of point masses $m_1, m_2, m_3, \ldots, m_n$, each gravitational force acts independently and uninfluenced by the presence of other bodies. The total gravitational force $\mathbf{F}_1$ on m_1 is the vector sum of the individual forces exerted by all other point masses:

$$\mathbf{F}_1 = \frac{Gm_2m_1}{r_{21}^2}\hat{\mathbf{r}}_{21} + \frac{Gm_3m_1}{r_{31}^2}\hat{\mathbf{r}}_{31} + \frac{Gm_4m_1}{r_{41}^2}\hat{\mathbf{r}}_{41} + \cdots = \sum_{i=2}^{n}\mathbf{F}_{1i}$$



Tutorial Diagram 3: (a) Vector displacement $r = r_2 - r_1$ between two point masses. (b) Total force on m_1 is the vector sum of attractions from m_2, m_3, and m_4.

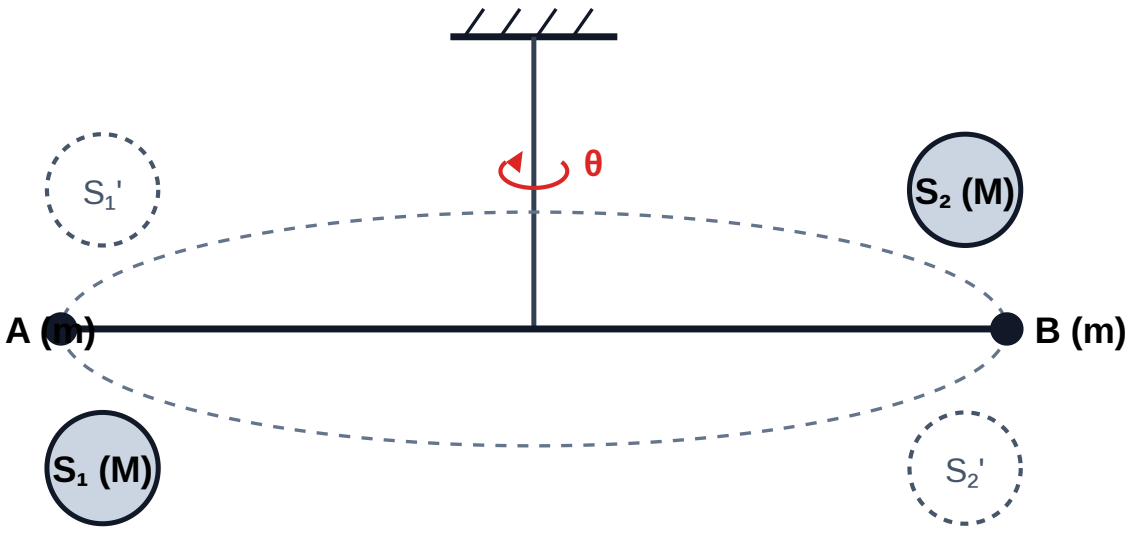
Newton's Shell Theorems (Extended Spherically Symmetric Bodies)

For extended objects of finite size, we must integrate the vector forces from all constituent point masses. For uniform spherical shells, calculus yields two fundamental results:

1. **Point Mass Outside a Spherical Shell:** The gravitational force of attraction between a hollow spherical shell of uniform density and a point mass situated *outside* it is exactly as if the entire mass of the shell were concentrated at its center. (Perpendicular force components from symmetric regions cancel out, leaving only the radial component along the line joining the point mass to the center.)
2. **Point Mass Inside a Spherical Shell:** The net gravitational force of attraction due to a hollow spherical shell of uniform density on a point mass situated anywhere *inside* it is **zero**. Forces exerted in various directions by different regions of the shell cancel each other completely. Note, however, that a spherical shell does *not* shield a body inside it from the gravitational attraction of external bodies outside the shell (gravitational shielding is impossible).

2.3 The Gravitational Constant (G) — Cavendish's Experiment

In 1798, English scientist **Henry Cavendish** experimentally determined the value of the universal gravitational constant G using a torsion balance apparatus.



Tutorial Diagram 4: Schematic of Cavendish's torsion balance. Large lead spheres S_1 and S_2 attract small spheres A and B on bar AB of length L, twisting the suspension wire by angle θ.

A light bar AB of length L with two small lead spheres (each of mass m) at its ends is suspended horizontally from a rigid support by a fine torsion wire. Two large lead spheres S_1 and S_2 (each of mass M) are brought close to the small spheres on opposite sides at a center-to-center distance d. The equal and opposite gravitational forces $F = \frac{GMm}{d^2}$ produce zero net translational force on the bar, but create a gravitational torque of magnitude $F \cdot L$.

The wire twists until its restoring torque $\tau\theta$ (where τ is the restoring couple per unit angle of twist and θ is the angle of twist in radians) balances the gravitational torque at equilibrium:

$$G\frac{Mm}{d^2}L = \tau\theta \implies G = \frac{\tau\theta d^2}{MmL}$$

The currently accepted value of the universal gravitational constant is:

$$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$$

2.4 Acceleration Due to Gravity of the Earth (g)

The Earth can be modeled as a sphere of radius R_E and mass M_E composed of concentric uniform spherical shells. For any point on or outside the Earth's surface, all shells attract a point mass m as if the entire mass M_E were concentrated at the Earth's center. On the surface of the Earth ($r = R_E$), the gravitational force is $F = \frac{GM_E m}{R_E^2}$. By Newton's second law ($F = mg$), the acceleration due to gravity on the Earth's surface is:

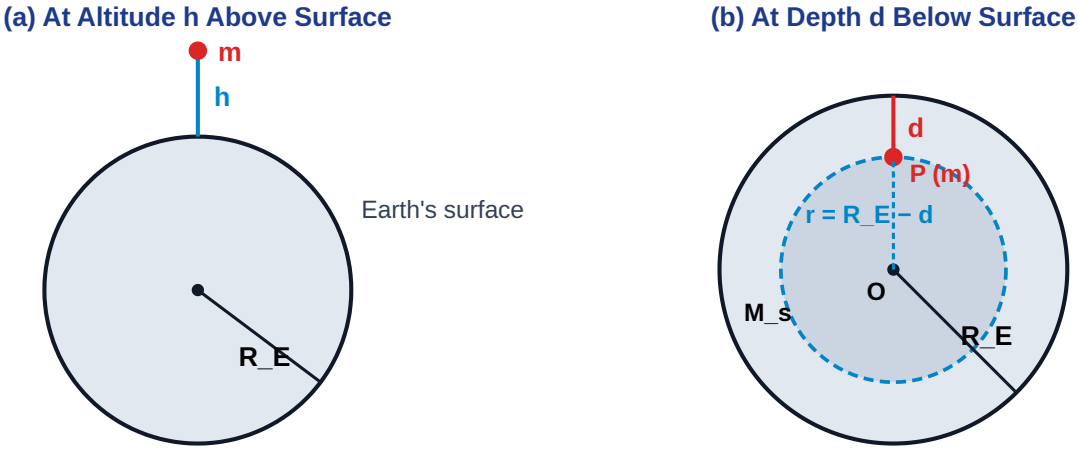
$$g = \frac{GM_E}{R_E^2}$$

In terms of the Earth's average uniform density ρ, since $M_E = \frac{4}{3}\pi R_E^3 \rho$, we can also write:

$$g = \frac{4}{3}\pi G R_E \rho$$

Because g and R_E were already known, Cavendish's measurement of G allowed scientists to compute $M_E = \frac{gR_E^2}{G} \approx 6.0 \times 10^{24}\,\mathrm{kg}$—hence the popular remark that *"Cavendish weighed the Earth."*

2.5 Variation of Acceleration Due to Gravity Above and Below Earth's Surface



Tutorial Diagram 5: (a) Point mass m at height h above Earth's surface. (b) Point mass m at depth d below Earth's surface; only the inner sphere of radius (R_E − d) and mass M_s exerts a net gravitational force.

1. Above the Surface of the Earth (at Altitude h)

At a height h above the Earth's surface, the distance from the Earth's center is $r = R_E + h$. The exact acceleration due to gravity $g(h)$ is:

$$g(h) = \frac{GM_E}{(R_E + h)^2} = \frac{g}{\left(1 + \frac{h}{R_E}\right)^2}$$

Binomial Approximation for Small Heights ($h \ll R_E$): When $h/R_E \ll 1$, expanding $\left(1 + \frac{h}{R_E}\right)^{-2} \approx 1 - \frac{2h}{R_E}$ gives:

$$g(h) \simeq g\left(1 - \frac{2h}{R_E}\right) \quad (\text{valid only for } h \ll R_E)$$

2. Below the Surface of the Earth (at Depth d)

Consider a point mass m in a mine at depth d below the surface, at distance $r = R_E - d$ from the center. The Earth can be divided into an outer spherical shell of thickness d and an inner solid sphere of radius $(R_E - d)$. By Newton's shell theorem, the outer shell exerts **zero** net force on m. Assuming the Earth has uniform density ρ, the mass M_s of the inner sphere is:

$$M_s = M_E \frac{(R_E - d)^3}{R_E^3} = M_E \frac{r^3}{R_E^3}$$

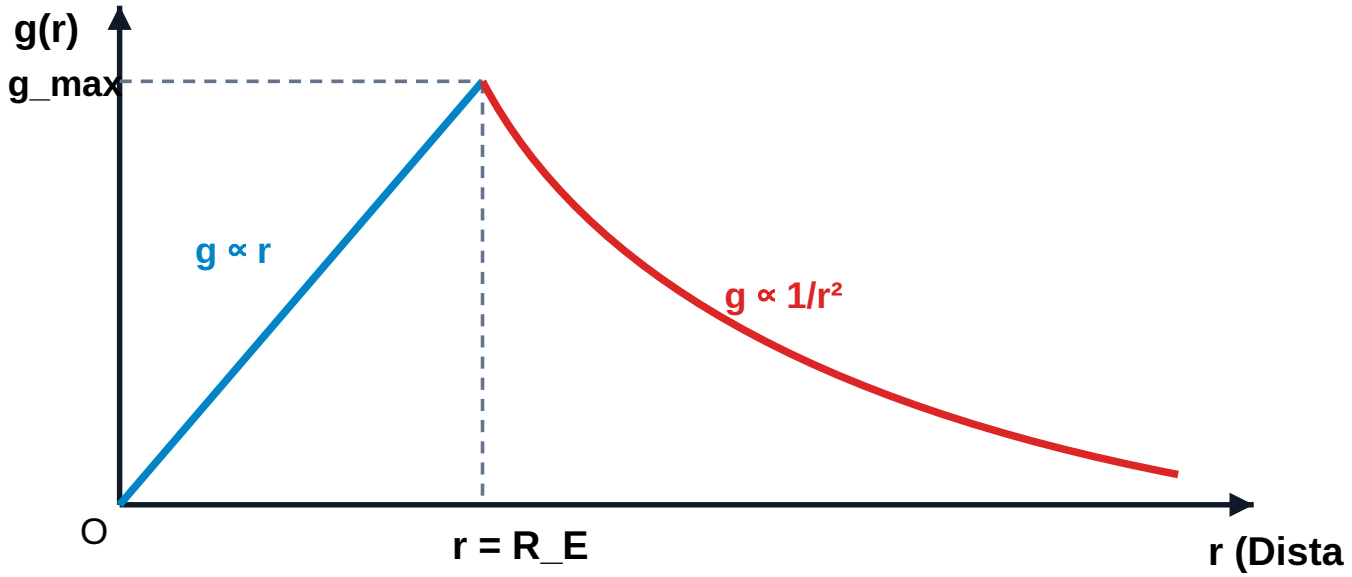
The gravitational force on m at depth d is therefore:

$$F(d) = \frac{GM_s m}{(R_E - d)^2} = \frac{GM_E m}{R_E^3}(R_E - d) = \frac{GM_E m}{R_E^3} r$$

Hence, the acceleration due to gravity at depth d is:

$$g(d) = \frac{F(d)}{m} = \frac{GM_E}{R_E^3} r = g\left(\frac{R_E - d}{R_E}\right) = g\left(1 - \frac{d}{R_E}\right)$$

Key Takeaway: Acceleration due to Earth's gravity is **maximum at its surface** ($r = R_E$). Inside the uniform Earth ($r < R_E$), $g \propto r$ (increases linearly from zero at the center to g at the surface). Outside the Earth ($r > R_E$), $g \propto \frac{1}{r^2}$.



Tutorial Diagram 6: Variation of acceleration due to gravity g(r) with distance r from the center of a uniformly dense spherical Earth.

2.6 Gravitational Potential Energy and Gravitational Potential

Gravitational force is a **conservative force** (work done is independent of the path taken). Near the surface of the Earth ($h \ll R_E$), where gravity is practically constant ($F = mg$), the work done in lifting a particle from height h_1 to h_2 is $W_{12} = mg(h_2 - h_1)$.

For arbitrary distances r_1 and r_2 ($r_2 > r_1 \geq R_E$) from the center of the Earth, the gravitational force varies as $F(r) = \frac{GM_E m}{r^2}$. The work done in lifting the particle radially from r_1 to r_2 is:

$$W_{12} = \int_{r_1}^{r_2} \frac{GM_E m}{r^2}\,\mathrm{d}r = -GM_E m\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = W(r_2) - W(r_1)$$

where the gravitational potential energy function at distance r is:

$$W(r) = -\frac{GM_E m}{r} + W_1$$

Here W_1 is the potential energy at infinity ($r \to \infty$). By convention, we choose $W_1 = 0$ at infinity. Thus, the **gravitational potential energy** V of two point masses m_1 and m_2 separated by distance r is:

$$V(r) = -\frac{Gm_1 m_2}{r} \quad (\text{with } V(\infty) = 0)$$

For an isolated system of multiple particles, by the superposition principle, the total gravitational potential energy is the algebraic sum of the potential energies of **all possible pairs** of constituent particles.

Gravitational Potential (U)

The **gravitational potential** at a point in a gravitational field is defined as the potential energy of a particle of *unit mass* placed at that point (i.e., work done in bringing a unit mass from infinity to that point):

$$U(r) = \frac{V(r)}{m} = -\frac{GM}{r} \quad (\text{SI Unit: J kg}^{-1})$$

2.7 Escape Speed (v_e)

Escape speed is the minimum initial speed required to project a body so that it escapes the gravitational pull of a planet and reaches infinity ($r \to \infty$).

Suppose a body of mass m is projected with initial speed V_i from a height h above the Earth's surface (distance $r = R_E + h$ from the center) and reaches infinity with final speed V_f. By conservation of mechanical energy (taking $W_1 = 0$ at infinity):

$$E_{\text{initial}} = E_{\text{final}} \implies \frac{1}{2}mV_i^2 - \frac{GmM_E}{R_E + h} = \frac{1}{2}mV_f^2 \geq 0$$

The minimum initial speed $(V_i)_{\min}$ corresponds to $V_f = 0$ at infinity:

$$\frac{1}{2}m(V_i^2)_{\min} = \frac{GmM_E}{R_E + h} \implies (V_i)_{\min} = \sqrt{\frac{2GM_E}{R_E + h}}$$

When projected from the **surface of the Earth** ($h = 0$), using $GM_E = gR_E^2$:

$$v_e = (V_i)_{\min} = \sqrt{\frac{2GM_E}{R_E}} = \sqrt{2gR_E} \approx 11.2\,\text{km}\,\text{s}^{-1}$$

- Escape speed is **independent of the mass** m of the projected body and **independent of the direction of projection** (provided it does not collide with the planet). It depends on the mass and radius of the planet and the height/location of projection.
- **Why the Moon Has No Atmosphere:** On the surface of the Moon, both surface gravity and radius are smaller than on Earth, yielding an escape speed of only $v_{e,\text{Moon}} \approx 2.3\,\text{km}\,\text{s}^{-1}$ (about five times smaller than Earth's). Gas molecules formed on the Moon's surface acquire thermal speeds greater than $2.3\,\text{km}\,\text{s}^{-1}$ and escape into space.

2.8 Earth Satellites: Orbital Speed and Time Period

Consider a satellite of mass m revolving around the Earth in a circular orbit of radius $r = R_E + h$ with orbital speed V. The required centripetal force is provided by Earth's gravitational attraction:

$$\frac{mV^2}{R_E + h} = \frac{GM_E m}{(R_E + h)^2}$$

$$V = \sqrt{\frac{GM_E}{R_E + h}}$$

Notice that as altitude h increases, the orbital speed V decreases. For a satellite orbiting very close to the Earth's surface ($h \approx 0$):

$$V_0 = V(h = 0) = \sqrt{\frac{GM_E}{R_E}} = \sqrt{gR_E} = \frac{v_e}{\sqrt{2}} \approx 7.92\,\text{km}\,\text{s}^{-1}$$

Time Period of Revolution (T): In one complete orbit, the satellite travels a circumference $2\pi(R_E + h)$ at speed V:

$$T = \frac{2\pi(R_E + h)}{V} = \frac{2\pi(R_E + h)^{3/2}}{\sqrt{GM_E}} \implies T^2 = \left(\frac{4\pi^2}{GM_E}\right)(R_E + h)^3 = k(R_E + h)^3$$

where $k = \frac{4\pi^2}{GM_E} \approx 10^{-13}\,\text{s}^2\,\text{m}^{-3}$. For a near-Earth satellite ($h \ll R_E$), using $R_E = 6400\,\text{km} = 6.4 \times 10^6\,\text{m}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$:

$$T_0 = 2\pi\sqrt{\frac{R_E}{g}} = 2\pi\sqrt{\frac{6.4 \times 10^6}{9.8}}\,\text{s} \approx 5075\,\text{s} \approx 84.6\,\text{minutes} \approx 85\,\text{minutes}$$

2.9 Mechanical Energy of an Orbiting Satellite

For a satellite of mass m in a circular orbit of radius $r = R_E + h$ around the Earth, its kinetic energy (K), potential energy (V, with zero reference at infinity), and total mechanical energy ($E = K + V$) are:

$$K = \frac{1}{2}mV^2 = +\frac{GM_E m}{2(R_E + h)}$$

$$V = -\frac{GM_E m}{R_E + h} = -2K$$

$$E = K + V = -\frac{GM_E m}{2(R_E + h)} = -K = \frac{V}{2}$$

- The total mechanical energy E of any bound (closed circular or elliptical) orbit is always **negative**. If $E \geq 0$, the object is unbound and escapes to infinity.
- In a circular orbit, $K = -E$ and $V = 2E$. When a satellite is moved to a higher circular orbit (larger r), its total energy E and potential energy V *increase* (become less negative), while its kinetic energy K and orbital speed *decrease*.
- In an **elliptical orbit** of semi-major axis a, both K and V vary from point to point, but the total mechanical energy remains constant and equals $E = -\frac{GM_E m}{2a}$.

3. Key Physical Quantities, Dimensions & Formulas

Physical Quantity	Symbol	Dimensions	SI Unit	Formula / Remarks
Universal Gravitational Constant	G	$[\mathrm{M}^{-1}\mathrm{L}^{3}\mathrm{T}^{-2}]$	$\mathrm{N\,m^2\,kg^{-2}}$	$6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ (Scalar)
Gravitational Force	$\mathbf{F}$	$[\mathrm{M}^{1}\mathrm{L}^{1}\mathrm{T}^{-2}]$	N	$\mathbf{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}}$ (Vector, attractive)
Gravitational Intensity (Field / Acceleration)	$\mathbf{E}$ or $\mathbf{g}$	$[\mathrm{M}^{0}\mathrm{L}^{1}\mathrm{T}^{-2}]$	$\mathrm{m\,s^{-2}}$ or $\mathrm{N\,kg^{-1}}$	$\mathbf{g} = -\frac{GM}{r^2}\hat{\mathbf{r}}$ (Vector)
Gravitational Potential Energy	$V(r)$	$[\mathrm{M}^{1}\mathrm{L}^{2}\mathrm{T}^{-2}]$	J	$V(r) = -\frac{GMm}{r}$ (Scalar; $V(\infty) = 0$)
Gravitational Potential	$U(r)$	$[\mathrm{M}^{0}\mathrm{L}^{2}\mathrm{T}^{-2}]$	$\mathrm{J\,kg^{-1}}$	$U(r) = -\frac{GM}{r}$ (Scalar)
Areal Velocity	$\frac{\Delta A}{\Delta t}$	$[\mathrm{M}^{0}\mathrm{L}^{2}\mathrm{T}^{-1}]$	$\mathrm{m^2\,s^{-1}}$	$\frac{\Delta A}{\Delta t} = \frac{L}{2m} = \text{constant}$

4. Problem-Solving Methods & Conceptual Connections

1. **Elliptical Orbit Extremes (Perihelion & Aphelion):** At the perihelion P ($r_P = a(1-e)$) and aphelion A ($r_A = a(1+e)$), the velocity vector $\mathbf{v}$ is strictly perpendicular to the position vector $\mathbf{r}$ ($\theta = 90^\circ$). Therefore, conservation of angular momentum simplifies directly to:
$$mv_P r_P = mv_A r_A \implies \frac{v_P}{v_A} = \frac{r_A}{r_P} = \frac{1+e}{1-e}$$
2. **When to Use Exact vs. Approximate Gravity Formulas:**
   - Use $g(h) = g\left(1 - \frac{2h}{R_E}\right)$ **only** when $h \ll R_E$ (typically $h \leq 5\%$ of $R_E \approx 320\,\mathrm{km}$).
   - If h is comparable to R_E (e.g., $h = R_E/2$, $h = R_E$), you **must** use the exact relation $g(h) = \frac{g}{(1+h/R_E)^2}$.
   - The depth formula $g(d) = g\left(1 - \frac{d}{R_E}\right)$ is **exact for all depths** $0 \leq d \leq R_E$ (assuming uniform density).
3. **Projectiles Reaching Comparable Heights (h Comparable to R_E):** Never use $v^2 = 2gh$ when initial speed is comparable to escape speed. Always apply conservation of mechanical energy:
$$\frac{1}{2}mv^2 - \frac{GM_Em}{R_E} = 0 - \frac{GM_Em}{R_E + h} \implies h = \frac{R_E}{\left(\frac{v_e^2}{v^2} - 1\right)}$$
4. **Speed Far Away (At Infinity) When Projected with $v > v_e$:** If a body is projected with initial speed $v > v_e$, it reaches infinity with speed v_f given by energy conservation:
$$\frac{1}{2}mv^2 - \frac{1}{2}mv_e^2 = \frac{1}{2}mv_f^2 \implies v_f = \sqrt{v^2 - v_e^2}$$
5. **Neutral Point Between Two Massive Bodies:** Between two fixed spheres of masses M_1 and M_2 separated by distance D, the gravitational forces cancel at a neutral point N at distance r from M_1 where $\frac{GM_1}{r^2} = \frac{GM_2}{(D-r)^2}$, so $r = \frac{\sqrt{M_1}}{\sqrt{M_1}+\sqrt{M_2}}D$. To transfer a projectile from the surface of M_1 to M_2, it only needs enough initial kinetic energy to just reach the neutral point N with $v_N \to 0$; beyond N, the attraction of M_2 pulls it in automatically.
6. **Weightlessness & Tidal Effects:**
   - An astronaut in an orbiting satellite experiences **weightlessness** not because $g = 0$, but because both the astronaut and the satellite are in continuous free fall with the same acceleration towards the Earth, making the normal reaction zero. (In a very large space station, small spatial variations in $\mathbf{g}$ could allow gravity detection.)
   - Although the Sun's gravitational pull on Earth is much stronger than the Moon's, the **tidal effect** depends on the differential gravitational pull across Earth's diameter ($\propto \frac{M}{r^3}$). Because the Moon is much closer to Earth than the Sun, the Moon's tidal effect is greater than the Sun's.

5. Comprehensive Worked Examples

Worked Example 1: Perihelion vs. Aphelion Speeds and Transit Times

Consider a planet moving in an elliptical orbit around the Sun S (refer to Tutorial Diagram 1a). Let the speed of the planet at the perihelion P be v_P and the Sun–planet distance SP be r_P. Relate $\{r_P, v_P\}$ to the corresponding quantities at the aphelion $\{r_A, v_A\}$. Will the planet take equal times to traverse the semi-elliptical paths BAC and CPB?

Solution:

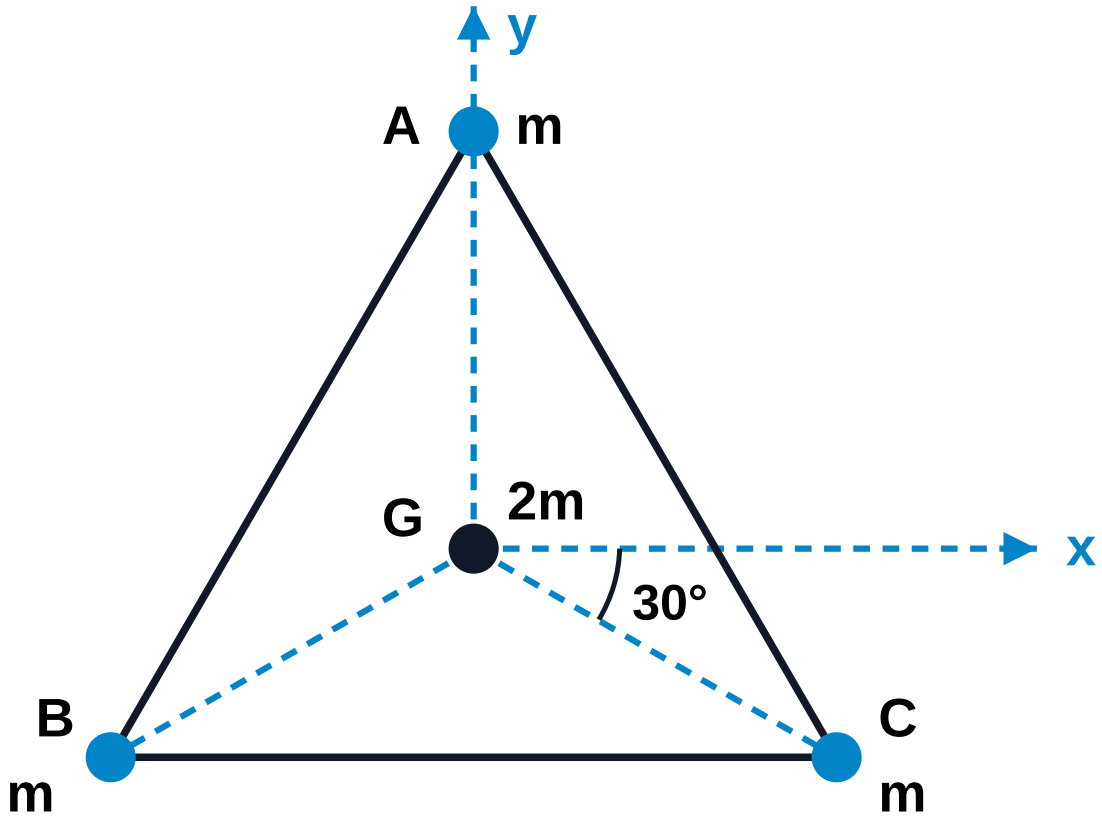
At both the perihelion P and aphelion A, the position vector $\mathbf{r}$ from the Sun and the velocity vector $\mathbf{v}$ of the planet are mutually perpendicular ($\theta = 90^\circ$). Thus, the magnitude of angular momentum of the planet (mass m_p) at P is $L_P = m_p r_P v_P$, and at A is $L_A = m_p r_A v_A$.

By conservation of angular momentum ($L_P = L_A$):
$$m_p r_P v_P = m_p r_A v_A \implies \frac{v_P}{v_A} = \frac{r_A}{r_P}$$
Since $r_A > r_P$, it follows that $v_P > v_A$ (the planet moves fastest at perihelion and slowest at aphelion).

To compare the transit times along paths BAC and CPB, consider the areas swept out by the radius vector from the Sun S. The area $SBAC$ (bounded by the right half of the ellipse BAC and the radius vectors SB and SC) is equal to the area of the right half-ellipse plus triangle SBC. The area $SBPC$ (bounded by the left half of the ellipse CPB and SB, SC) is equal to the left half-ellipse minus triangle SBC. Thus, $\mathrm{Area}(SBAC) > \mathrm{Area}(SBPC)$. By Kepler's Second Law (equal areas are swept in equal times), the planet takes a **longer time to traverse** BAC **than** CPB.

Worked Example 2: Superposition of Gravitational Forces on an Equilateral Triangle

Three equal point masses of $m\,\text{kg}$ each are fixed at the vertices A, B, and C of an equilateral triangle ABC. A mass $2m$ is placed at the centroid G of the triangle, with $AG = BG = CG = 1\,\text{m}$.
(a) What is the resultant gravitational force acting on the mass $2m$ at G?
(b) What is the resultant gravitational force on $2m$ if the mass at vertex A is doubled to $2m$?



Tutorial Diagram 7: Three masses m at vertices A, B, C of an equilateral triangle and mass 2m at centroid G.

Solution:

(a) Choose coordinate axes with origin at G, GA along the $+y$-axis ($\hat{\mathbf{j}}$), and the x-axis horizontal. Then GB and GC make angles of 30° below the $-x$ and $+x$ axes, respectively. Using $AG = BG = CG = 1\,\text{m}$, the individual gravitational forces on $2m$ are:

$$\mathbf{F}_{GA} = \frac{Gm(2m)}{1^2}\hat{\mathbf{j}} = 2Gm^2\hat{\mathbf{j}}$$

$$\mathbf{F}_{GB} = 2Gm^2(-\hat{\mathbf{i}}\cos 30^\circ - \hat{\mathbf{j}}\sin 30^\circ)$$

$$\mathbf{F}_{GC} = 2Gm^2(+\hat{\mathbf{i}}\cos 30^\circ - \hat{\mathbf{j}}\sin 30^\circ)$$

By the principle of superposition, the resultant force $\mathbf{F}_R$ is:

$$\mathbf{F}_R = \mathbf{F}_{GA} + \mathbf{F}_{GB} + \mathbf{F}_{GC} = 2Gm^2\hat{\mathbf{j}} - 4Gm^2\sin 30^\circ\hat{\mathbf{j}} = 2Gm^2\hat{\mathbf{j}} - 2Gm^2\hat{\mathbf{j}} = \mathbf{0}$$

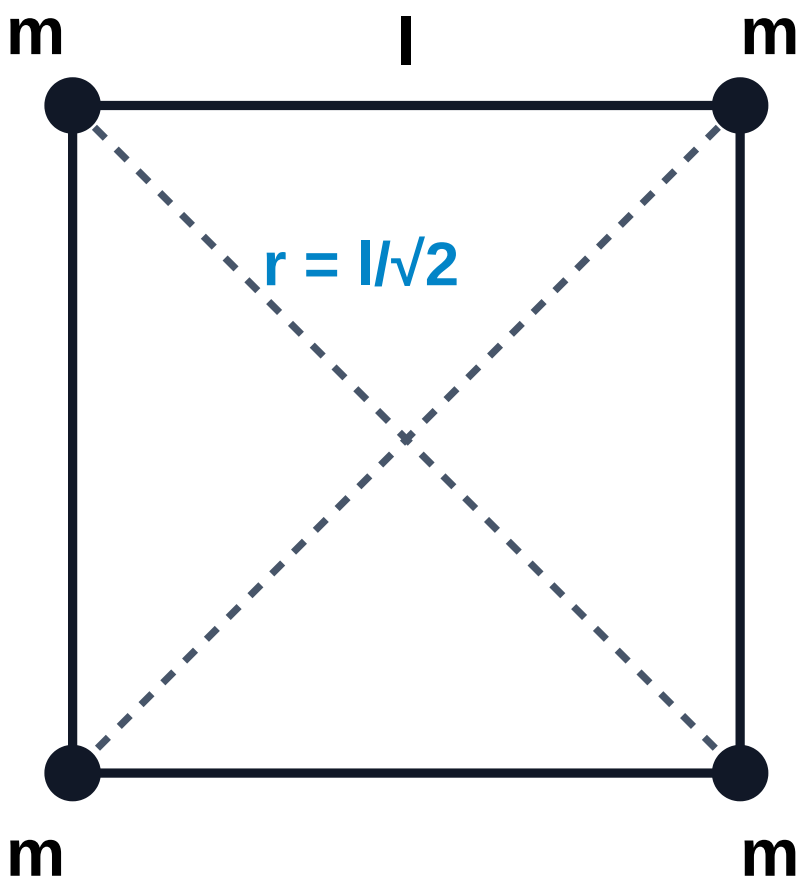
(This also follows immediately from three-fold rotational symmetry.)

(b) When the mass at vertex A is doubled to $2m$, $\mathbf{F}'_{GA} = \frac{G(2m)(2m)}{1^2}\hat{\mathbf{j}} = 4Gm^2\hat{\mathbf{j}}$, while $\mathbf{F}_{GB} + \mathbf{F}_{GC} = -2Gm^2\hat{\mathbf{j}}$ remains unchanged. Therefore:

$$\mathbf{F}'_R = 4Gm^2\hat{\mathbf{j}} - 2Gm^2\hat{\mathbf{j}} = 2Gm^2\hat{\mathbf{j}}$$

Worked Example 3: Gravitational Potential Energy and Potential of a Square Configuration

Four identical particles, each of mass m, are placed at the vertices of a square of side l. Find the total gravitational potential energy of the four-particle system and the gravitational potential at the center of the square.



Tutorial Diagram 8: Four particles of mass m at the corners of a square of side l.

Solution:

1. Total Potential Energy of the System: For $N = 4$ particles, there are $\frac{4\times3}{2} = 6$ pairs of particles: 4 adjacent side pairs separated by distance l, and 2 diagonal pairs separated by distance $\sqrt{2}l$. Summing over all 6 pairs:

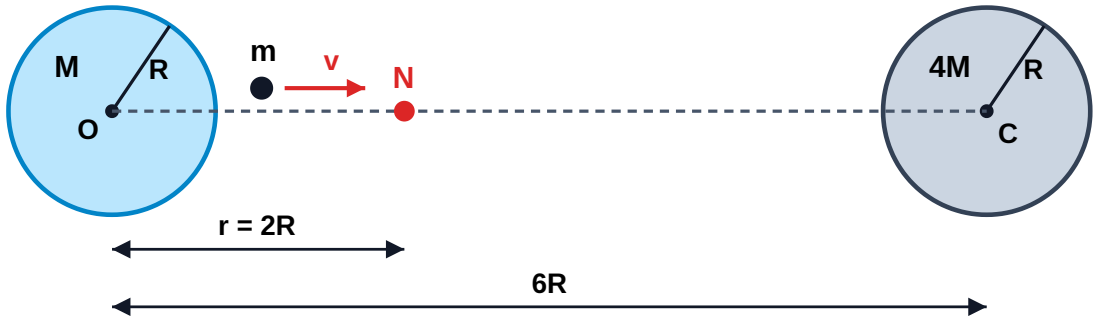
$$W = -4\frac{Gm^2}{l} - 2\frac{Gm^2}{\sqrt{2}l} = -\frac{2Gm^2}{l}\left(2 + \frac{1}{\sqrt{2}}\right) = -5.41\frac{Gm^2}{l}$$

2. Gravitational Potential at the Center: The distance of each corner from the center of the square is $r = \frac{\sqrt{2}l}{2} = \frac{l}{\sqrt{2}}$. Adding the scalar potentials due to all four masses:

$$U = 4 \times \left(-\frac{Gm}{r}\right) = 4 \times \left(-\frac{Gm}{l/\sqrt{2}}\right) = -4\sqrt{2}\frac{Gm}{l}$$

Worked Example 4: Minimum Projection Speed Between Two Fixed Solid Spheres

Two uniform solid spheres of equal radii R, but masses M and $4M$, have a center-to-center separation of $6R$ and are held fixed. A projectile of mass m is launched from the surface of the sphere of mass M directly towards the center of the second sphere ($4M$). Obtain an expression for the minimum speed v of the projectile so that it reaches the surface of the second sphere.



Tutorial Diagram 9: Projectile of mass m launched from sphere M towards sphere 4M across neutral point N.

Solution:

Step 1: Locate the Neutral Point N: Let N be at distance $ON = r$ from the center O of sphere M (and distance $6R - r$ from the center C of sphere $4M$) where the two opposing gravitational attractions cancel:

$$\frac{GMm}{r^2} = \frac{G(4M)m}{(6R-r)^2} \implies (6R-r)^2 = 4r^2 \implies 6R - r = \pm 2r$$

Taking the root between the spheres gives $r = 2R$ (since $r = -6R$ lies outside the region between the spheres).

Step 2: Apply Conservation of Mechanical Energy: The projectile only needs enough initial speed v at the surface of M (where its distance from O is R and from C is $6R - R = 5R$) to barely reach the neutral point N ($v_N \to 0$, where its distance from O is $2R$ and from C is $6R - 2R = 4R$). Beyond N, the stronger pull of $4M$ accelerates it to the second sphere:

$$E_i = E_N \implies \frac{1}{2}mv^2 - \frac{GMm}{R} - \frac{4GMm}{5R} = 0 - \frac{GMm}{2R} - \frac{4GMm}{4R}$$

$$\frac{1}{2}v^2 - \frac{GM}{R} - \frac{4GM}{5R} = -\frac{GM}{2R} - \frac{GM}{R}$$

$$\frac{1}{2}v^2 = \frac{GM}{R}\left(\frac{4}{5} - \frac{1}{2}\right) = \frac{3GM}{10R} \implies v = \sqrt{\frac{3GM}{5R}}$$

Worked Example 5: Mass of Mars and Length of the Martian Year

The planet Mars has two moons, Phobos and Deimos.
(i) Phobos has an orbital period of $7\,\text{hours}\ 39\,\text{minutes}$ and an orbital radius of $9.4 \times 10^3\,\text{km}$. Calculate the mass of Mars.
(ii) Assume that Earth and Mars move in circular orbits around the Sun, with the Martian orbital radius being 1.52 times the orbital radius of the Earth. What is the length of the Martian year in days?

Solution:

(i) Convert the orbital period and radius of Phobos into SI units:

$$T = 7\,\text{h}\ 39\,\text{min} = 459\,\text{min} = 459 \times 60\,\text{s} = 2.754 \times 10^4\,\text{s}, \quad R = 9.4 \times 10^3\,\text{km} = 9.4 \times 10^6\,\text{m}$$

From Kepler's third law applied to Phobos orbiting Mars of mass M_m:

$$T^2 = \frac{4\pi^2}{GM_m}R^3 \implies M_m = \frac{4\pi^2 R^3}{GT^2} = \frac{4 \times (3.14)^2 \times (9.4 \times 10^6)^3}{6.67 \times 10^{-11} \times (459 \times 60)^2} = 6.48 \times 10^{23}\,\text{kg}$$

(ii) Applying Kepler's third law to Mars and Earth orbiting the Sun, with $\frac{R_{MS}}{R_{ES}} = 1.52$ and $T_E = 365$ days:

$$\frac{T_M^2}{T_E^2} = \left(\frac{R_{MS}}{R_{ES}}\right)^3 \implies T_M = (1.52)^{3/2} \times 365\,\text{days} \approx 1.874 \times 365\,\text{days} \approx 684\,\text{days}$$

Worked Example 6: Weighing the Earth by Two Independent Methods

Given $g = 9.81\,\mathrm{m\,s^{-2}}$, radius of the Earth $R_E = 6.37 \times 10^6\,\mathrm{m}$, distance to the Moon $R = 3.84 \times 10^8\,\mathrm{m}$, time period of the Moon's revolution $T = 27.3\,\mathrm{days}$, and $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$, calculate the mass of the Earth M_E in two different ways.

Solution:
Method 1 (From Terrestrial Acceleration Due to Gravity):

$$M_E = \frac{gR_E^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.97 \times 10^{24}\,\mathrm{kg}$$

Method 2 (From Moon's Orbital Motion via Kepler's Third Law):

$$M_E = \frac{4\pi^2 R^3}{GT^2} = \frac{4 \times (3.14)^2 \times (3.84 \times 10^8)^3}{6.67 \times 10^{-11} \times (27.3 \times 24 \times 60 \times 60)^2} = 6.02 \times 10^{24}\,\mathrm{kg}$$

Both independent methods yield values in remarkable agreement (differing by less than 1%).

Worked Example 7: Energy Required for Orbit Transfer of a Satellite

A $400\,\mathrm{kg}$ satellite is in a circular orbit of radius $2R_E$ about the Earth. How much energy is required to transfer it to a circular orbit of radius $4R_E$? What are the changes in its kinetic and potential energies? (Take $g = 9.81\,\mathrm{m\,s^{-2}}$ and $R_E = 6.37 \times 10^6\,\mathrm{m}$).

Solution:
In a circular orbit of radius r, the total mechanical energy is $E = -\frac{GM_E m}{2r}$. Thus, for initial radius $r_i = 2R_E$ and final radius $r_f = 4R_E$:

$$E_i = -\frac{GM_E m}{2(2R_E)} = -\frac{GM_E m}{4R_E}, \qquad E_f = -\frac{GM_E m}{2(4R_E)} = -\frac{GM_E m}{8R_E}$$

The energy required to transfer the satellite is the increase in total mechanical energy $\Delta E = E_f - E_i$:

$$\Delta E = E_f - E_i = \frac{GM_E m}{8R_E} = \left(\frac{GM_E}{R_E^2}\right)\frac{mR_E}{8} = \frac{gmR_E}{8} = \frac{9.81 \times 400 \times 6.37 \times 10^6}{8} = 3.13 \times 10^9\,\mathrm{J}$$

Since $K = -E$ and $V = 2E$ for circular orbits:

- Change in Kinetic Energy: $\Delta K = K_f - K_i = -\Delta E = -3.13 \times 10^9\,\mathrm{J}$ (kinetic energy decreases).
- Change in Potential Energy: $\Delta V = V_f - V_i = 2\Delta E = +6.25 \times 10^9\,\mathrm{J}$ (potential energy increases).

6. NEET UG Exam-Style Practice Questions with Worked Solutions

Note: The following are original exam-aligned practice questions designed strictly for NEET UG preparation.

Practice Question 1

Single-Correct MCQ | Kepler's Third Law

Suppose there existed a planet that went around the Sun twice as fast as the Earth (i.e., its orbital time period is half that of the Earth). What would be its orbital size (semi-major axis) compared to that of the Earth ($1\,\mathrm{AU}$)?

(A) $0.50\,\mathrm{AU}$

(B) $0.63\,\mathrm{AU}$

(C) $0.79\,\mathrm{AU}$

(D) $0.25\,\mathrm{AU}$

Correct Answer: (B) $0.63\,\mathrm{AU}$
Worked Solution: Given $T_p = \frac{1}{2}T_e$ and $a_e = 1\,\mathrm{AU}$. By Kepler's Third Law ($T^2 \propto a^3$):

$$\frac{a_p}{a_e} = \left(\frac{T_p}{T_e}\right)^{2/3} = \left(\frac{1}{2}\right)^{2/3} = (0.25)^{1/3} \approx 0.63\,\mathrm{AU}$$

Practice Question 2

Single-Correct MCQ | Variation of g with Altitude

A body weighs $63\,\mathrm{N}$ on the surface of the Earth. What is the gravitational force on it due to the Earth at a height equal to half the radius of the Earth ($h = R_E/2$)?

(A) $31.5\,\mathrm{N}$

(B) $28\,\mathrm{N}$

(C) $42\,\mathrm{N}$

(D) $15.75\,\mathrm{N}$

Correct Answer: (B) $28\,\mathrm{N}$
Worked Solution: Since $h = R_E/2$ is comparable to R_E, we must use the exact formula for gravity at height h:

$$g(h) = \frac{g}{\left(1 + \frac{h}{R_E}\right)^2} = \frac{g}{\left(1 + \frac{1}{2}\right)^2} = \frac{g}{\left(\frac{3}{2}\right)^2} = \frac{4}{9}g$$

Therefore, the weight at $h = R_E/2$ is $W(h) = \frac{4}{9}W_{\text{surface}} = \frac{4}{9} \times 63\,\mathrm{N} = 28\,\mathrm{N}$.

Practice Question 3

Single-Correct MCQ | Variation of g with Depth

Assuming the Earth to be a sphere of uniform mass density, how much would a body weigh halfway down to the center of the Earth ($d = R_E/2$) if it weighed $250\,\mathrm{N}$ on the surface?

(A) $500\,\mathrm{N}$

(B) $62.5\,\mathrm{N}$

(C) $125\,\mathrm{N}$

(D) $250\,\mathrm{N}$

Correct Answer: (C) $125\,\mathrm{N}$
Worked Solution: At depth $d = R_E/2$, the acceleration due to gravity is:

$$g(d) = g\left(1 - \frac{d}{R_E}\right) = g\left(1 - \frac{1}{2}\right) = \frac{g}{2}$$

Hence, the weight of the body at depth $d = R_E/2$ is $W(d) = \frac{1}{2} \times 250\,\mathrm{N} = 125\,\mathrm{N}$.

Practice Question 4

Single-Correct MCQ | Energy Conservation & Maximum Height

A rocket is fired vertically with a speed of $5\,\mathrm{km\,s^{-1}}$ from the Earth's surface. How far from the Earth's surface does the rocket go before returning to the Earth? (Given: $M_E = 6.0 \times 10^{24}\,\mathrm{kg}$, $R_E = 6.4 \times 10^6\,\mathrm{m}$, $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$).

(A) $1.6 \times 10^6\,\mathrm{m}$

(B) $8.0 \times 10^6\,\mathrm{m}$

(C) $1.28 \times 10^6\,\mathrm{m}$

(D) $3.2 \times 10^6\,\mathrm{m}$

Correct Answer: (A) $1.6 \times 10^6\,\mathrm{m}$
Worked Solution: By conservation of mechanical energy between the surface ($r = R_E, v = 5 \times 10^3\,\mathrm{m\,s^{-1}}$) and maximum height h ($r = R_E + h, v = 0$):

$$\frac{1}{2}mv^2 - \frac{GM_Em}{R_E} = -\frac{GM_Em}{R_E + h} \implies \frac{GM_E}{R_E + h} = \frac{GM_E}{R_E} - \frac{v^2}{2} = \frac{2GM_E - v^2R_E}{2R_E}$$

$$R_E + h = \frac{2GM_ER_E}{2GM_E - v^2R_E} = \frac{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 6.4 \times 10^6}{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24} - (5 \times 10^3)^2 \times 6.4 \times 10^6} = \frac{5.12 \times 10^{21}}{8.0 \times 10^{14} - 1.6 \times 10^{14}} = 8.0 \times 10^6\,\mathrm{m}$$

Therefore, height above the Earth's surface is $h = 8.0 \times 10^6\,\mathrm{m} - 6.4 \times 10^6\,\mathrm{m} = 1.6 \times 10^6\,\mathrm{m}$.

Practice Question 5

Single-Correct MCQ | Projection Above Escape Speed

The escape speed of a projectile on the Earth's surface is $11.2\,\mathrm{km\,s^{-1}}$. A body is projected out with thrice this speed ($v_i = 3v_e$). What is the speed of the body far away from the Earth ($r \to \infty$)? Ignore the presence of the Sun and other planets.

(A) $22.4\,\mathrm{km\,s^{-1}}$

(B) $31.68\,\mathrm{km\,s^{-1}}$

(C) $33.6\,\mathrm{km\,s^{-1}}$

(D) $11.2\sqrt{3}\,\mathrm{km\,s^{-1}}$

Correct Answer: (B) $31.68\,\mathrm{km\,s^{-1}}$

Worked Solution: Using conservation of mechanical energy with $\frac{GM_Em}{R_E} = \frac{1}{2}mv_e^2$:

$$\frac{1}{2}mv_i^2 - \frac{1}{2}mv_e^2 = \frac{1}{2}mv_f^2 \implies v_f = \sqrt{v_i^2 - v_e^2} = \sqrt{(3v_e)^2 - v_e^2} = \sqrt{8}\,v_e = 2\sqrt{2} \times 11.2\,\mathrm{km\,s^{-1}} \approx 31.68\,\mathrm{km\,s^{-1}}$$

Practice Question 6

Single-Correct MCQ | Binding Energy of a Satellite

A satellite of mass $200\,\mathrm{kg}$ orbits the Earth at a height of $400\,\mathrm{km}$ above the surface. How much energy must be expended to rocket the satellite out of the Earth's gravitational influence? (Given: $M_E = 6.0 \times 10^{24}\,\mathrm{kg}$, $R_E = 6.4 \times 10^6\,\mathrm{m}$, $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$).

(A) $5.9 \times 10^9\,\mathrm{J}$

(B) $11.8 \times 10^9\,\mathrm{J}$

(C) $2.95 \times 10^9\,\mathrm{J}$

(D) $6.25 \times 10^9\,\mathrm{J}$

Correct Answer: (A) $5.9 \times 10^9\,\mathrm{J}$

Worked Solution: The orbital radius is $r = R_E + h = 6.4 \times 10^6\,\mathrm{m} + 0.4 \times 10^6\,\mathrm{m} = 6.8 \times 10^6\,\mathrm{m}$. The total mechanical energy of the orbiting satellite is $E = -\frac{GM_Em}{2(R_E+h)}$. To just escape to infinity ($E_f = 0$), the energy required is $\Delta E = 0 - E = |E|$:

$$\Delta E = \frac{GM_Em}{2(R_E+h)} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 200}{2 \times 6.8 \times 10^6} \approx 5.89 \times 10^9\,\mathrm{J} \approx 5.9 \times 10^9\,\mathrm{J}$$

Practice Question 7

Single-Correct MCQ | Gravitational Neutral Point

A rocket is fired from the Earth towards the Sun. At what distance from the Earth's center is the net gravitational force on the rocket zero? (Given: Mass of the Sun $M_S = 2 \times 10^{30}\,\mathrm{kg}$, mass of the Earth $M_E = 6 \times 10^{24}\,\mathrm{kg}$, Earth–Sun orbital distance $D = 1.5 \times 10^{11}\,\mathrm{m}$).

(A) $2.59 \times 10^8\,\mathrm{m}$

(B) $5.18 \times 10^8\,\mathrm{m}$

(C) $1.5 \times 10^9\,\mathrm{m}$

(D) $3.84 \times 10^8\,\mathrm{m}$

Correct Answer: (A) $2.59 \times 10^8\,\mathrm{m}$

Worked Solution: Let x be the distance from the Earth's center where the gravitational pulls of Earth and Sun cancel:

$$\frac{GM_Em}{x^2} = \frac{GM_Sm}{(D-x)^2} \implies \frac{D-x}{x} = \sqrt{\frac{M_S}{M_E}} = \sqrt{\frac{2 \times 10^{30}}{6 \times 10^{24}}} = \sqrt{\frac{1}{3} \times 10^6} \approx 577.35$$

$$x = \frac{D}{578.35} = \frac{1.5 \times 10^{11}\,\mathrm{m}}{578.35} \approx 2.59 \times 10^8\,\mathrm{m}$$

Practice Question 8

Single-Correct MCQ | Head-On Collision of Two Stars

Two stars, each of one solar mass ($M = 2 \times 10^{30}\,\text{kg}$) and radius $R = 10^4\,\text{km} = 10^7\,\text{m}$, are approaching each other for a head-on collision. When they are at a separation of $d = 10^9\,\text{km} = 10^{12}\,\text{m}$, their speeds are negligible. With what speed v (each) do they collide? (Assume the stars remain undistorted until collision and $G = 6.67 \times 10^{-11}\,\text{N}\,\text{m}^2\,\text{kg}^{-2}$).

(A) $1.3 \times 10^6\,\text{m}\,\text{s}^{-1}$

(B) $2.58 \times 10^6\,\text{m}\,\text{s}^{-1}$

(C) $5.16 \times 10^6\,\text{m}\,\text{s}^{-1}$

(D) $6.45 \times 10^5\,\text{m}\,\text{s}^{-1}$

Correct Answer: (B) $2.58 \times 10^6\,\text{m}\,\text{s}^{-1}$
Worked Solution: By symmetry, in the center-of-mass frame both equal-mass stars have equal speeds v at collision (when their centers are separated by $2R = 2 \times 10^7\,\text{m}$). Conserving total mechanical energy of the two-star system:

$$-\frac{GM^2}{d} = 2\left(\frac{1}{2}Mv^2\right) - \frac{GM^2}{2R} \implies v^2 = GM\left(\frac{1}{2R} - \frac{1}{d}\right)$$

Since $d = 10^{12}\,\text{m} \gg 2R = 2 \times 10^7\,\text{m}$, we can neglect $\frac{1}{d}$ compared to $\frac{1}{2R}$:

$$v \approx \sqrt{\frac{GM}{2R}} = \sqrt{\frac{6.67 \times 10^{-11} \times 2 \times 10^{30}}{2 \times 10^7}} = \sqrt{6.67 \times 10^{12}} \approx 2.58 \times 10^6\,\text{m}\,\text{s}^{-1}$$

Practice Question 9

Single-Correct MCQ | Equilibrium & Potential Between Spheres

Two heavy spheres each of mass $100\,\text{kg}$ and radius $0.10\,\text{m}$ are placed $1.0\,\text{m}$ apart (center-to-center) on a horizontal table. What are the net gravitational force (per unit mass) and gravitational potential at the midpoint of the line joining their centers, and what is the nature of equilibrium of a particle placed there along the line joining the centers?

(A) Force $= 0$; Potential $= -2.67 \times 10^{-8}\,\text{J}\,\text{kg}^{-1}$; Unstable equilibrium

(B) Force $= 0$; Potential $= -1.33 \times 10^{-8}\,\text{J}\,\text{kg}^{-1}$; Stable equilibrium

(C) Force $= 2.67 \times 10^{-8}\,\text{N}\,\text{kg}^{-1}$; Potential $= 0$; Unstable equilibrium

(D) Force $= 0$; Potential $= -2.67 \times 10^{-8}\,\text{J}\,\text{kg}^{-1}$; Stable equilibrium

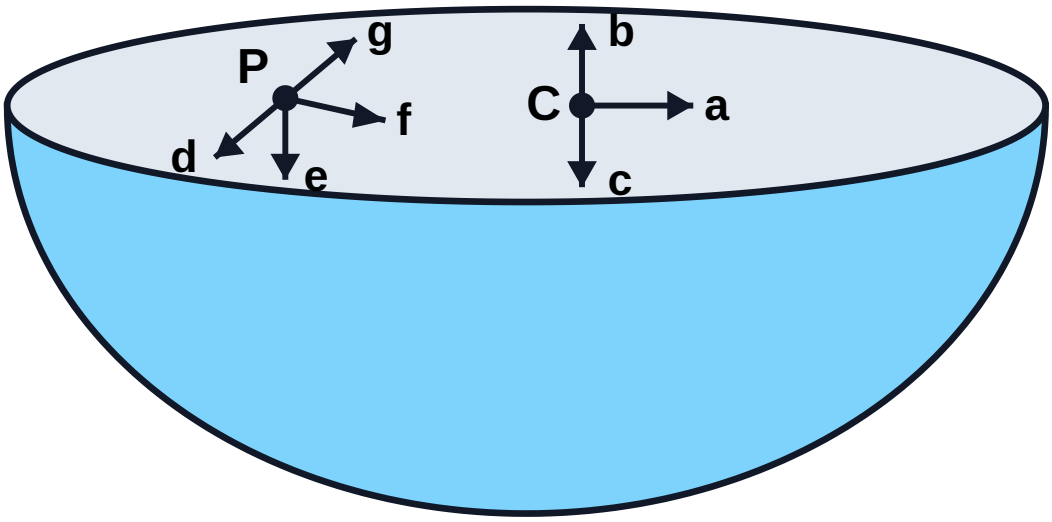
Correct Answer: (A) Force $= 0$; Potential $= -2.67 \times 10^{-8}\,\text{J}\,\text{kg}^{-1}$; Unstable equilibrium
Worked Solution: At the midpoint ($r = 0.5\,\text{m}$ from each center), the equal and opposite gravitational pulls cancel, so the net gravitational force is **zero**. The scalar gravitational potentials add:

$$U = -\frac{GM}{r} - \frac{GM}{r} = -\frac{2 \times 6.67 \times 10^{-11} \times 100}{0.5} = -2.67 \times 10^{-8}\,\text{J}\,\text{kg}^{-1}$$

If the test particle is displaced slightly along the line joining the centers towards either sphere, the attraction of the closer sphere exceeds that of the farther sphere, pulling the particle further away from the midpoint. Hence, the equilibrium is **unstable**.

Practice Question 10 Single-Correct MCQ | Hemispherical Shell & Shell Theorem

Consider a hemispherical shell of uniform mass density with its flat circular face horizontal at the top and its curved bowl downward, as shown in the diagram below. What are the directions of the gravitational intensity (field) at the center C of the circular face and at an arbitrary off-center point P on the flat circular face?



Tutorial Diagram 10: Hemispherical shell of uniform mass density showing candidate field directions at center C and arbitrary point P on the equatorial plane.

(A) At C: arrow c; At P: arrow e

(B) At C: arrow b; At P: arrow g

(C) At C: zero; At P: arrow d

(D) At C: arrow c; At P: arrow f

Correct Answer: (A) At C: arrow c; At P: arrow e
Worked Solution: Complete the uniform spherical shell by adding an upper hemisphere. Inside a complete uniform spherical shell, the gravitational intensity is zero everywhere ($\mathbf{g}_{\text{lower}} + \mathbf{g}_{\text{upper}} = \mathbf{0} \implies \mathbf{g}_{\text{lower}} = -\mathbf{g}_{\text{upper}}$). At the center C, symmetry dictates that the lower hemisphere pulls straight down along arrow c. At any point P on the flat circular plane of symmetry dividing the two hemispheres, horizontal components of $\mathbf{g}_{\text{lower}}$ and $\mathbf{g}_{\text{upper}}$ would be identical by mirror symmetry, so their sum can be zero only if the horizontal component vanishes! Thus, at every point P on the flat circular face, the gravitational intensity due to the lower hemisphere is strictly perpendicular to the plane, pointing vertically downward along arrow e.

Practice Question 11

Multi-Statement Selection | Comet in Elliptical Orbit

A comet orbits the Sun in a highly elliptical orbit (neglect any mass loss of the comet). Which of the following physical quantities remain **constant** throughout its orbit?
I. Linear speed
II. Angular speed
III. Angular momentum
IV. Kinetic energy
V. Potential energy
VI. Total mechanical energy

(A) I, II, and III only

(B) III and VI only

(C) III, IV, V, and VI only

(D) II, III, and VI only

Correct Answer: (B) III and VI only
Worked Solution: Because the Sun's gravitational pull is a central conservative force, only the **angular momentum (III)** and **total mechanical energy (VI)** of the comet remain constant throughout the elliptical orbit. Because the Sun–comet distance r varies continuously, the linear speed, angular speed, kinetic energy, and potential energy all vary from point to point.

Practice Question 12

Assertion–Reason | NEET UG Format

Assertion (A): A hollow spherical shell of uniform density cannot shield a body placed inside it from the gravitational influence of nearby external matter.
Reason (R): Gravitational force is independent of the intervening medium and obeys the principle of superposition, where external masses exert their gravitational pull independently of the shell.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) Both (A) and (R) are false.

Correct Answer: (A) Both (A) and (R) are true, and (R) is the correct explanation of (A).
Worked Solution: Unlike a hollow metallic conductor (where mobile charges redistribute to cancel external electric fields inside), matter has only one sign of gravitational mass and the gravitational force due to external bodies acts independently by the superposition principle. While the shell's *own* net gravitational force on an interior particle is zero, external bodies continue to attract the interior particle unhindered.

Practice Question 13

Assertion–Reason | Satellite Launch Energy

Assertion (A): The energy required to launch an orbiting satellite out of Earth's gravitational influence is *less* than the energy required to project a stationary object at the same height (as the satellite) out of Earth's gravitational influence.
Reason (R): An orbiting satellite at height h already possesses positive kinetic energy $K = \frac{GM_Em}{2(R_E+h)}$, so its total energy is $-\frac{GM_Em}{2(R_E+h)}$, whereas a stationary object at the same height has a lower total energy of $-\frac{GM_Em}{R_E+h}$.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A) Both (A) and (R) are true, and (R) is the correct explanation of (A).
Worked Solution: To escape to infinity ($E_f = 0$), an orbiting satellite at radius $r = R_E + h$ requires additional energy $\Delta E_{\text{sat}} = \frac{GM_Em}{2(R_E+h)}$, whereas a stationary object at the same height ($K = 0$) requires $\Delta E_{\text{stat}} = \frac{GM_Em}{R_E+h} = 2\Delta E_{\text{sat}}$. Hence, the orbiting satellite requires half as much additional energy.

Practice Question 14 Single-Correct MCQ | Saturn's Orbital Distance

A Saturn year is 29.5 times the Earth year. How far is Saturn from the Sun if the Earth is $1.50 \times 10^8\ \text{km}$ away from the Sun?

(A) $1.43 \times 10^9\ \text{km}$

(B) $7.78 \times 10^8\ \text{km}$

(C) $2.87 \times 10^9\ \text{km}$

(D) $4.50 \times 10^9\ \text{km}$

Correct Answer: (A) $1.43 \times 10^9\ \text{km}$

Worked Solution: Using Kepler's Third Law ($R \propto T^{2/3}$):

$$R_{\text{Saturn}} = R_{\text{Earth}} \left(\frac{T_{\text{Saturn}}}{T_{\text{Earth}}} \right)^{2/3} = (1.50 \times 10^8\ \text{km}) \times (29.5)^{2/3} = 1.50 \times 10^8 \times 9.547\ \text{km} \approx 1.43 \times 10^9\ \text{km}$$

7. Common NEET UG Pitfalls & Revision Checklist

Pitfall 1: Confusing Distance from Surface (h) with Distance from Center (r)

In all orbital, potential, and exact gravity equations, $r = R_E + h$ is measured from the **center of the Earth**. If a question states "at a height equal to R_E", then $r = 2R_E$, so $g' = g/4$ (not g). Conversely, if a question states "at a distance $2R_E$ from the center", then $r = 2R_E$ directly.

Pitfall 2: Misusing the Binomial Approximation $g(h) \approx g(1 - 2h/R_E)$

Never apply $g(h) = g(1 - 2h/R_E)$ for large altitudes such as $h = R_E/2$ or $h = R_E$ (for $h = R_E/2$, the binomial formula gives $g(h) = 0$, which is completely wrong; the exact value is $\frac{4}{9}g$).

Pitfall 3: Double-Counting Pairs in System Potential Energy

For a system of N particles, there are $\frac{N(N-1)}{2}$ unique pairs. For a triangle ($N = 3$), there are 3 pairs; for a square or tetrahedron ($N = 4$), there are 6 pairs. Do not multiply the potential of a single vertex by N without dividing by 2!

Pitfall 4: Sign Errors in Satellite Energy Relations

Remember the exact sign relationships for a circular orbit: $K > 0$, $V < 0$, $E < 0$, with $E = -K = \frac{V}{2}$. Thus, the total energy of an orbiting satellite is the **negative of its kinetic energy** and **half of its potential energy**.

Quick Revision Checklist for NEET UG

- Kepler's Second Law $\left(\frac{\Delta A}{\Delta t} = \frac{L}{2m} = \text{constant}\right)$ holds for *any* central force; linear momentum is not conserved for a planet alone, but angular momentum and total mechanical energy are conserved.
- At perihelion P and aphelion A: $v_P r_P = v_A r_A$, with $r_P = a(1 - e)$ and $r_A = a(1 + e)$.
- Inside a uniform solid sphere ($r < R_E$): $g(r) = g\frac{r}{R_E} = g\left(1 - \frac{d}{R_E}\right)$. Inside a uniform hollow spherical shell: $g = 0$ and potential $U = -\frac{GM}{R} = \text{constant}$.
- Escape speed from Earth's surface: $v_e = \sqrt{2gR_E} = \sqrt{2}\,V_0 \approx 11.2\ \text{km s}^{-1}$ (independent of projectile mass and launch angle).
- Near-Earth satellite ($h \ll R_E$): $V_0 = \sqrt{gR_E} \approx 7.92\ \text{km s}^{-1}$ and $T_0 = 2\pi\sqrt{\frac{R_E}{g}} \approx 85\ \text{minutes}$.
- In space (weightlessness), blood rushes to the upper body, causing a **swollen face**, headache, and orientational problems (not swollen feet).

8. Concise Final Recap

Gravitation is a universal, attractive, central, and conservative inverse-square force governed by $F = \frac{Gm_1m_2}{r^2}$. It explains both terrestrial free fall ($g = \frac{GM_E}{R_E^2}$) and celestial mechanics through Kepler's three laws. Earth's gravitational acceleration is maximum on its surface and decreases both with altitude h and depth d. Bound gravitational systems (planets and satellites) always possess negative total mechanical energy $E = -\frac{GMm}{2a}$, while supplying energy to make $E \geq 0$ allows a body to escape to infinity with minimum launch speed $v_e = \sqrt{\frac{2GM_E}{R_E}}$.